

Handle Control

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Note

Handle control requires two Microbits, and the micro-compact handle must be equipped.

Controller Expansion Pack Address: <https://github.com/YahboomTechnology/GHBitLib>. The factory sample source code has already been imported, so there's no need to import it again. For import instructions, please refer to: "2. Introduction to Graphic Programming -> [Import Lastbit Exclusive Expansion Pack] -> [Import in Empty Project]"

1. Control Principle

The Micro:bit has a built-in **nRF processor** with a **2.4GHz RF antenna**, which can be used to achieve communication between two Microbits.

2. Precautions

- The handle needs to be connected to No. 7 batteries.
- The left S2 toggle switch is the power switch. It must be turned on before using handle control. (Set to ON position)
- The right S3 toggle switch is the vibration switch. Decide whether to turn it on based on your personal needs.

3. Program Burning

1. Connect the Microbit mainboard to the computer using a data cable. At this point, you will see a MICROBIT drive appear on your computer. Here is one example, the drive letter is G.



2. Car side burning: Copy or send "**3.Handle_Control_Car_Side.hex**" to the Microbit drive. After successful burning, the **5*5 LED matrix** will display "T".
3. Handle side burning: Two programs are provided for the handle side:
3.Handle_Control_Handle_Side(No_Direction).hex and
3.Handle_Control_Handle_Side(Show_Direction).hex. After burning, it will display . You can copy or send any program to the Microbit drive. **The version without direction display is more sensitive.**



4. Button and Function Description

Handle Side Version Without Direction Display

Button Description	Function Description
Joystick Up	Car Forward
Joystick Down	Car Backward
Joystick Left	Car Turn Left
Joystick Right	Car Turn Right
Joystick Press	Turn Off Lights
B2 (Green)	Green Light On, Robotic Arm Moves Up
B1 (Red)	Red Light On, Robotic Arm Moves Left
B3 (Blue)	Blue Light On, Robotic Arm Moves Down
B4 (Yellow)	Yellow Light On, Robotic Arm Moves Right
A Button	Control Robotic Arm Claw to Grip

Button Description	Function Description
B Button	Control Robotic Arm Claw to Release
S2 (OFF ON)	Handle Power Switch - Set to (ON) when in use
S3 (Software Button)	Vibration Switch (When set to Button on the right side, pressing the right button will vibrate and the buzzer will sound)

Handle Side Version With Direction Display

Button Description	Function Description
Joystick Up	Car Forward, Display Up Arrow
Joystick Down	Car Backward, Display Down Arrow
Joystick Left	Car Turn Left, Display Left Arrow
Joystick Right	Car Turn Right, Display Right Arrow
Joystick Press	Turn Off Lights, Display "X"
B2 (Green)	Green Light On, Robotic Arm Moves Up
B1 (Red)	Red Light On, Robotic Arm Moves Left
B3 (Blue)	Blue Light On, Robotic Arm Moves Down
B4 (Yellow)	Yellow Light On, Robotic Arm Moves Right
A Button	Control Robotic Arm Claw to Grip
B Button	Control Robotic Arm Claw to Release
S2 (OFF ON)	Handle Power Switch - Set to (ON) when in use
S3 (Software Button)	Vibration Switch (When set to Button on the right side, pressing the right button will vibrate and the buzzer will sound)